Ghosting Suppression With the Joint Subchannel BP Image Reconstruction for Nonuniform Sampling SAR

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***Abstract*— In recent years, the backprojection (BP) algorithm has shown good potentials in synthetic aperture radar (SAR) focusing with nonideal conditions, such as nonideal trajectory and nonuniform sampling. However, when applying the original BP algorithm directly to periodic nonuniform sampling data, ghostings can be introduced into the imaging results, leading to image quality distortion. To address this issue, a novel reconstruction method based on joint subchannel BP images is proposed. We first analyze the causes and properties of Doppler grating lobes and BP ghostings in the periodic nonuniform signals. Based on the analysis, a reconstruction process of the joint subchannel BP images is established to achieve the purpose of suppressing ghosting. The proposed method can be smoothly combined with the ground Cartesian BP (GCBP) algorithm to realize high-computational efficiency. Finally, the simulation and measured data are presented to verify the effectiveness of the algorithm.**

***Index Terms*— Backprojection (BP), ghostings, nonuniform sampling, synthetic aperture radar (SAR).**

I. Introduction

A SYNTHETIC aperture radar (SAR) [1], [2], [3] is a well-proven imaging technique for remote sensing of the Earth. With the emergence of high-resolution and wide-width imaging requirements in the current development of SAR, the acquisition method of SAR signal is no longer limited to uniform sampling, and the processing techniques of nonuniform sampling data have gradually become a research hotspot in distributed SAR [4] and staggered SAR [5]. For instance, the distributed SAR uses the variable PRF schemes to prevent the periodic appearance of the coinciding samples and improve the conditions of the reconstruction process. The staggered SAR adopts a variable PRF scheme to eliminate the limited vision distance and achieve the purpose of increasing the width of mapping in the case of high-resolution SAR. Besides, the signal received by the multichannel SAR system [22], [23] can also be regarded as a periodic nonuniform sampling signal.

However, the nonuniform sampling will introduce some problems: irregular spectrum grating lobes and matching filter failure. The traditional standard frequency-domain imaging algorithms, such as the chirp scaling algorithm (CSA) [6], polar format algorithm (PFA) [7], and the typical Omega-K [8], [9], [10], cannot achieve good imaging performance in nonuniform SAR systems due to the matching filter failure. As an alternative, the standard time-domain imaging algorithm, the backprojection algorithm (BPA) [11], [12], can be utilized to effectively avoid the issue of matched filter failure. As a cost, backprojection (BP) will introduce a high-computational burden. For this reason, several fast BP algorithms [13], namely, the fast factorized BP (FFBP) algorithm, have been developed to reduce computational complexity. However, the coherent fusion of subaperture (SA) images in these fast BP algorithms relies on image-domain interpolation operations, which may not be practical in practical applications [14], [15]. In [16], the 2-D nonuniform FFT 2-D NUFFT) operation is employed in FFBP as a replacement for the interpolation operation. This utilization of 2-D NUFFT helps enhance the accuracy and speed of SA image fusion. Furthermore, the Cartesian factorized BP (CFBP) algorithms [17], [18], [19] have been proposed, where the imaging plane is established in Cartesian coordinates. In addition, the ground Cartesian BP (GCBP) [20], [21] algorithm based on "spectrum compression" is proposed to enhance the imaging accuracy and efficiency.

Although the BP algorithms mentioned above can effectively avoid the problem of matching filter failure, they are still unable to suppress the influence of azimuth irregular grating lobes. In [28], two effective reconstruction methods, NUFFT [24], [25] and Jenq's method [26], are introduced to suppress the irregular spectrum grating lobes. The NUFFT can also be utilized to correct the along-track error in near-space SAR [29]. In addition, the filter group algorithm [27] can be employed to suppress the irregular spectrum grating lobes by dividing the nonuniform data into multiple subchannel data for reconstruction. However, the filter group algorithm can only eliminate the grating lobes within the baseband, and

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the grating lobes outside the baseband will still be retained. Some methods [30], [31] based on sparse regulation schemes have been massively investigated for similar baseband external grating lobe suppression topics. However, these methods generally have problems, such as high-computational complexity and poor real-time performance. With the increase in scene complexity, their convergence accuracy and convergence speed will be greatly affected.

In this article, we are committed to finding an effective and efficient image-domain reconstruction method to suppress the grating lobes introduced by nonuniform sampling. First, the Doppler spectrum of the periodic nonuniform sampling SAR signal is analyzed in depth based on the nonuniform sampling model. Second, the problem of BP algorithm in a nonuniform SAR system is discussed and analyzed, i.e., the ghosts in BP results under nonuniform sampling. To address this issue, we propose a reconstruction method based on joint subchannel BP images. In this proposed algorithm, the mapping relationship between the BP image and the Doppler spectrum is skillfully excavated, and the reconstruction model in the joint subchannel BP images is established. Based on the reconstruction model, the reconstruction process of the joint subchannel BP image is realized. Compared with those algorithms [30], [31], the proposed algorithm in this article has a stable reconstruction model without iteration and optimization, and the speed and accuracy of the reconstruction processing are reliable. Finally, the simulation experiments and measured experiments in a nonuniform SAR system are carried out to verify the above analysis and discussion.

This article is organized as follows. In Section II, a detailed description of the problem is given. The proposed algorithm is described in detail in Section III. In Section IV, the SAR simulation experiments and measured experiments based on nonuniform sampling are provided. The conclusions are presented in Section V.

II. Problem Description

As mentioned above, nonuniform sampling has inherent advantages in distributed SAR and staggered SAR scenarios. In the following content, we delve into the topic of nonuniform sampling mode in more detail.

A. Nonuniform Sampling Mode

Fig. 1 gives the nonuniform sampling model, in which we can see that $t_n = nT + r_nT$, and the expression of the nonuniform sampling signal can be expressed as follows:

$$x_s(t_n) = x(t) \cdot s(t) \tag{1}$$

where $x(t)$ is a continuous-time signal and $s(t)$ is a sampling function, which is given by

$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT - r_nT). \tag{2}$$

Equation (2) is the sampling function. T is the nominal sampling period, and r_n is the timing offset measured in percentage of the nominal sampling period T. The periodic nonuniform sampling is more common in practice, and its

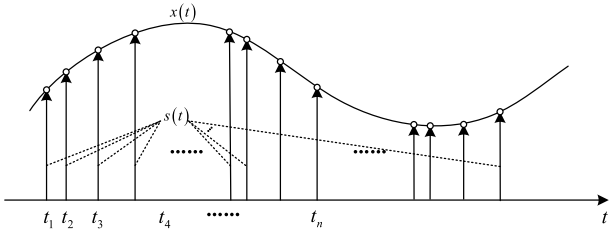


Fig. 1. Nonuniform sampling model.

sampling form is analyzed in the next. Let $n = kM + m$, where k ranges from $-\infty$ to ∞. Then, (2) can be further given by

$$s(t) = \sum_{m=0}^{M-1} \sum_{k=-\infty}^{\infty} \delta(t - (k \cdot M + m)T - r_mT) \tag{3}$$

where M is the period of the nonuniform sampling and $r_m = r_{kM+m}$. The second summation part of (3) is taken out separately and recorded as follows:

$$s_m(t) = \sum_{k=-\infty}^{\infty} \delta(t - (k \cdot M + m)T - r_mT). \tag{4}$$

Equation (2) can be simplified as $s(t) = \sum_{m=0}^{M-1} s_m(t)$. Obviously, the signal of (4) is a periodic impact sequence with a period of MT. Then, its Fourier series representation is

$$s_m(t) = \sum_{l=-\infty}^{\infty} a_{m,l} \exp(jl\Omega t) \tag{5}$$

where $\Omega = 2\pi/(MT)$, and the specific value of the Fourier coefficient $a_{m,l}$ can be obtained by the following integral:

$$a_{m,l} = \frac{1}{T_M} \int_{-\frac{T_M}{2}}^{\frac{T_M}{2}} s_m(t) e^{-jl\Omega t} dt = 1/T_M \cdot e^{-jl\Omega(r_m+m)T} \tag{6}$$

where $T_M = M \cdot T$. Substituting (6) into (5), we can obtain

$$s_m(t) = \frac{1}{T_M} \sum_{l=-\infty}^{\infty} \exp(jl\Omega(t - (r_m + m)T)). \tag{7}$$

Let $t' = t - (r_m + m)T$, and substituting it into (7), we have

$$s_m\left(t' + (r_m + m)T\right) = \frac{1}{T_M} \sum_{l=-\infty}^{\infty} \exp\left(jl\Omega t'\right) = s_{\text{ref}}\left(t'\right). \tag{8}$$

Then, the Fourier transform of (8) can be obtained

$$S_{\text{ref}}(j\omega) = \frac{1}{T_M} \sum_{l=-\infty}^{\infty} \delta(\omega - l\Omega) \tag{9}$$

where ω is the digital-frequency, which is closely combined with sampling. According to the time shift characteristics of Fourier transform, the Fourier transform of (7) can be further given by

$$S_m(j\omega) = \frac{1}{T_M} \sum_{l=-\infty}^{\infty} \delta(\omega - l\Omega) \cdot \exp(-j\omega(r_m + m)T). \tag{10}$$

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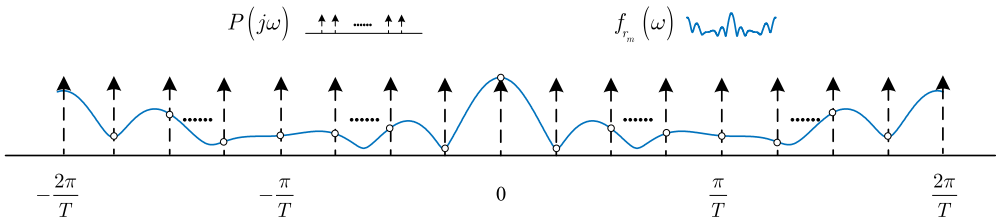


Fig. 2. Schematic of $P(j\omega)$ and $f_{r_m}(\omega)$ within two times the baseband.

Combining (3) and (10), we have

$$S(j\omega)=\frac{1}{T_M}\sum_{l=-\infty}^{\infty}\delta(\omega-l\Omega)\cdot\sum_{m=0}^{M-1}\exp(-j\omega(r_m+m)T). \quad (11)$$

In this article, (11) is deeply analyzed and rewritten as follows:

$$S(j\omega)=P(j\omega)\cdot f_{r_m}(\omega) \quad (12)$$

where

$$P(j\omega)=S_{\text{ref}}(j\omega),\quad f_{r_m}(\omega)=\sum_{m=0}^{M-1}\exp(-j\omega(r_m+m)T) \quad (13)$$

where $P(j\omega)$ is the first spectrum modulation function and $f_{r_m}(\omega)$ is the second spectrum modulation function. Fig. 2 gives the schematic of the $P(j\omega)$ and $f_{r_m}(\omega)$ within two times the baseband, namely, $(-2\pi/T, 2\pi/T)$. As can be seen from Fig. 2, within the baseband $(-\pi/T, \pi/T)$, the effective gain value of $S(j\omega)$, i.e., energy accumulation point, appears at the frequency $\omega_l = l\cdot(2\pi/MT)$ $(l\in[-M/2:M/2])$, that is, M gain points, and the existence of these gain points is the main reason for the existence of grating lobes in the spectrum, which will be discussed in detail in the later analysis.

Combining (13) with Fig. 2, (12) can be further rewritten as follows:

$$\begin{aligned}S(j\omega)&=1/T_M\sum_{l=-\infty}^{\infty}\sigma_l\cdot\delta(\omega-l\Omega)\\ \sigma_l&=f_{r_m}(l\Omega)=\sum_{m=0}^{M-1}\exp(-jl\Omega(r_m+m)T).\end{aligned} \quad (14)$$

Coefficients σ_l are related to the sampling structure of the signal. When $r_m=0$, the signal is uniformly collected, and we have

$$f_{r_m=0}(\omega)=f(\omega)=\sin\left(\omega\frac{MT}{2}\right)\Big/\sin\left(\frac{\omega T}{2}\right). \quad (15)$$

Fig. 3 gives the schematic of the $P(j\omega)$ and $f(\omega)$ within two times the baseband $(-2\pi/T, 2\pi/T)$. It can be seen from Fig. 3 that there is only one effective gain point of $S(j\omega)$ within the baseband $(-\pi/T, \pi/T)$, and the gain point is located at $\omega=0$. This is the reason why uniform sampling signals have no grating lobes.

Therefore, when the signal is uniformly sampled, $S(j\omega)$ degrades to $\tilde{S}(j\omega)$, as follows:

$$\tilde{S}(j\omega)=1/T\sum_{l=-\infty}^{\infty}\delta(\omega-l\cdot M\Omega). \quad (16)$$

In summary, whether grating lobes appear in the signal spectrum depends on the structure of the $f_{r_m}(\omega)$.

B. Nonuniform Sampling SAR Signal

In a nonuniform sampling SAR system, the expression of the 2-D SAR echo can be given by

$$\begin{aligned}x(t_r,t_n)=&\text{rect}\left(\frac{t_r-2R(t_n)/c}{T_p}\right)\cdot\exp\left(j\pi\gamma\left(t_r-\frac{2R(t_n)}{c}\right)^2\right)\\&\cdot W_a(t_n)\cdot\exp\left(-j\frac{4\pi}{\lambda}R(t_n)\right)\end{aligned} \quad (17)$$

where t_r is the fast time, T_p is the pulsewidth, and $\lambda=c/f_c$ is the wavelength. γ is the frequency rate in the range, and t_n is the nonuniform slow time. rect$(\cdot)$ and $W_a(\cdot)$ are the range window function and azimuth window function, respectively. After range compression, we have

$$\begin{aligned}xc_1(t_r,t_n)=&W_a(t_n)\cdot\sin c\left(\frac{1}{B}\left(t_r-2R(t_n)/c\right)\right)\\&\cdot\exp\left(-j\frac{4\pi}{\lambda}R(t_n)\right).\end{aligned} \quad (18)$$

Also, the specific expression of the slant range history is given by

$$R(t_n)=\sqrt{(R_B-y_i)^2+(v\cdot t_n-x_i)^2} \quad (19)$$

where R_B is the nearest range from the SAR trajectory to the center of the scene. Referring to (1), we can further get

$$xc_1(t_r,t_n)=xc(t_r,t)\cdot s(t). \quad (20)$$

By performing the nonuniform discrete Fourier transform (NUDFT) on (20), the Doppler expression is obtained

$$\begin{aligned}Xc_1(t_r,j\omega)&=Xc(t_r,j\omega)\otimes S(j\omega)\\&=Xc(t_r,j\omega)\otimes\frac{1}{T_M}\sum_{l=-\infty}^{\infty}\sigma_l\cdot\delta(\omega-l\Omega)\\&=\frac{1}{T_M}\sum_{l=-\infty}^{\infty}Xc(\omega-l\Omega)\cdot f_{r_m}(jl\Omega).\end{aligned} \quad (21)$$

To explain (21) in more detail, we first discuss and analyze $f_{r_m}(\omega)$. First, we consider that when $f_{r_m}(\omega)$ does not exist, the form of the Doppler spectrum in (21) degenerates to

$$\begin{aligned}Xc_2(t_r,j\omega)&=Xc(t_r,j\omega)\otimes P(j\omega)\\&=\left(1/T_M\right)\sum_{l=-\infty}^{\infty}Xc(t_r,j(\omega-l\Omega)).\end{aligned} \quad (22)$$

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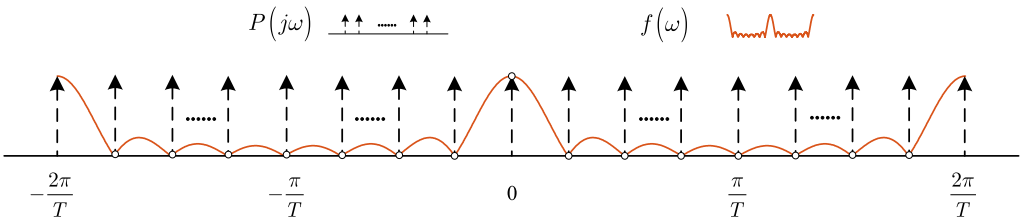


Fig. 3. Schematic of $P(j\omega)$ and $f(\omega)$ within two times the baseband.

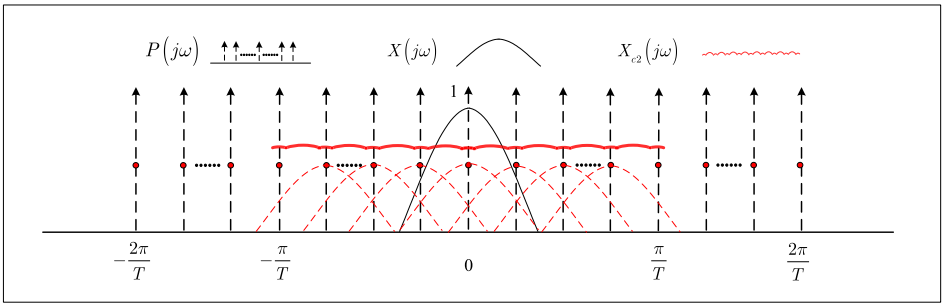


Fig. 4. Schematic of $Xc_2(t_r, j\omega)$ within two times the baseband.

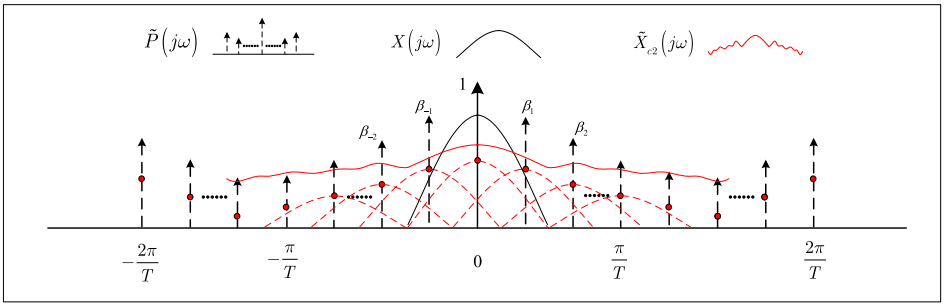


Fig. 5. Schematic of $\tilde{X}c_2(t_r, j\omega)$ within two times the baseband.

Fig. 4 gives the schematic of the $Xc_2(t_r, j\omega)$ within two times the baseband $(-2\pi/T, 2\pi/T)$. However, it should be pointed out that the effective Doppler bandwidth of $Xc_2(t_r, j\omega)$ is $(-\pi/MT, \pi/MT)$. In most cases, $Xc_2(t_r, j\omega)$ is aliasing, as shown in Fig. 4, and we cannot accurately extract the Doppler spectrum component $Xc(t_r, j\omega)$ from the aliasing $Xc_2(t_r, j\omega)$. In practical engineering, due to some constraints, the periodic assumption in (4) is approximated, so that $P(j\omega)$ degenerates into the following form:

$$\tilde{P}(j\omega) = \left(1/T_M\right) \sum_{l=-M/2}^{M/2} \beta_l \cdot \delta(\omega - l\Omega) \tag{23}$$

where β_l is a complex modulation function, whose expression is hard to obtain. However, the overall trend of β_l decreases with the increase of $|l|$ in $l \in [-M/2 : M/2]$ from the simulation results when r_m monotonically changes. Fig. 5 shows the spectrum results $Xc_2(t_r, j\omega)$ under $\tilde{P}(j\omega)$ modulation when r_m monotonically changes.

Recalling to (21), $P(j\omega)$ and $f_{r_m}(\omega)$ modulate the Doppler spectrum $Xc(t_r, j\omega)$ together to obtain $Xc_1(t_r, j\omega)$, and the schematic of this theory processing is shown in Fig. 6. Compared with Fig. 4, it can be seen that by introducing the synergistic effect of $f_{r_m}(\omega)$, the spectrum result of the signal is greatly improved. In practical engineering, $\tilde{P}(j\omega)$ replaces the role of $P(j\omega)$, and $\tilde{X}c_1(t_r, j\omega)$ spectrum schematic is shown in Fig. 7. It can be found that the grating lobe is further suppressed, and the result of $\tilde{X}c_1(t_r, j\omega)$ is closer to $Xc(t_r, j\omega)$. Combined with the above analysis, it can be concluded that for the bandlimited signal, the approximation degree of $\tilde{X}c_1(t_r, j\omega)$ and $Xc(t_r, j\omega)$ depends on the number of gain points of $S(j\omega) = P(j\omega) \cdot f_{r_m}(\omega)$ in the baseband $(-\pi/T, \pi/T)$. Considering the uniform sampling, the modulation function $S(j\omega)$ becomes $\tilde{S}(j\omega)$, and $\tilde{X}c_1(t_r, j\omega) = (1/T) \cdot Xc(t_r, j\omega)$ can be obtained, which is the classical relation of the spectral relationship before and after the signal is sampled. Clearly and intuitively, Fig. 8 shows the Doppler spectrum of uniformly sampling signal, in which one can see that no grating lobe is left in the baseband $(-\pi/T, \pi/T)$.

C. Analysis of Nonuniform Signal in BP Algorithm

It is well known that the core equation of the BP algorithm is given by

$$i(x_i, y_i) = \sum_{t_n} xc_1(t_r, t_n) \cdot \exp\left(j\frac{4\pi}{\lambda} R(t_n, x_i, y_i)\right). \tag{24}$$

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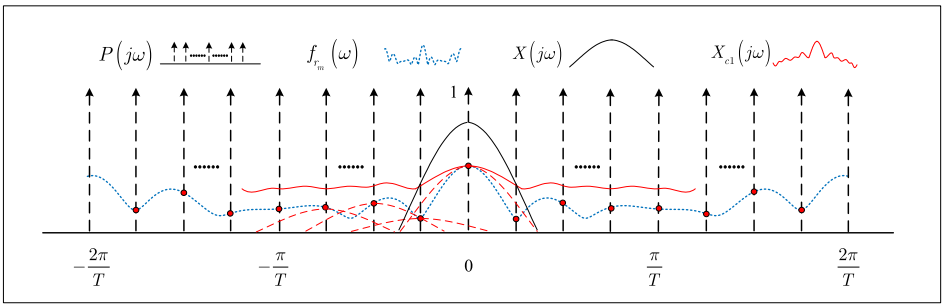


Fig. 6. Schematic of $Xc_1(t_r, j\omega)$ within two times the baseband.

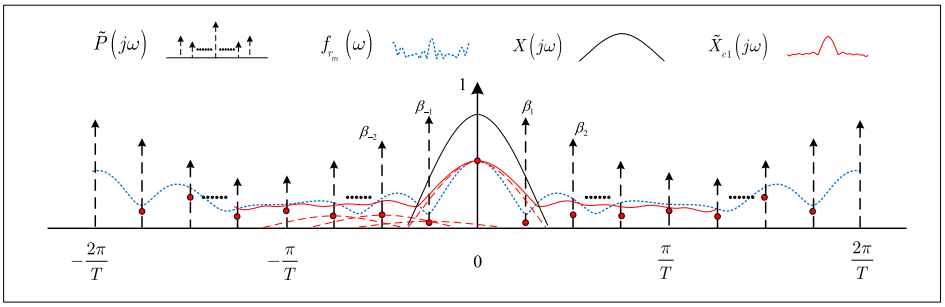


Fig. 7. Schematic of $\tilde{X}c_1(t_r, j\omega)$ within two times the baseband.

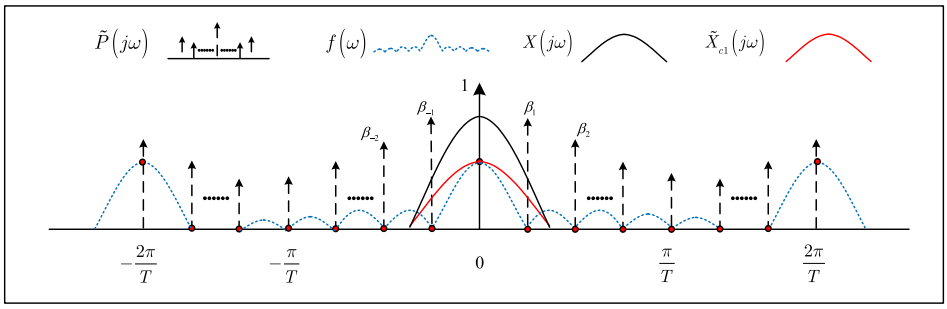


Fig. 8. In the case of uniform sampling: schematic of $\tilde{X}c_1(t_r, j\omega)$ within two times the baseband.

For subsequent analysis, let

$$yc_1(t_r, t_n) = xc_1(t_r, t_n)\exp\left(j\frac{4\pi}{\lambda}R(t_n, x_i, y_i)\right) \tag{25}$$

by applying NUDFT directly on (25) and take the value at $\omega = 0$, and we have

$$i(x_i, y_i) = \left(\sum_{t_n} yc_1(t_r, t_n)\exp(-j\omega t_n)\right)\Big|_{\omega=0}. \tag{26}$$

Then, we can further obtain

$$i(x_i, y_i) = Yc_1(t_r, j\omega)|_{\omega=0}. \tag{27}$$

Combining the analysis of (21) above, we can obtain

$$\begin{aligned} Yc_1(t_r, j\omega) &= Yc(t_r, j\omega) \otimes \left\{P(j\omega)\cdot f_{r_m}(\omega)\right\} \\ &= Yc(t_r, j\omega) \otimes \sum_{l=-\infty}^{\infty}\alpha_l \cdot \delta(\omega - l\Omega). \end{aligned} \tag{28}$$

However, there are M grating lobes in $Yc_1(t_r, j\omega)$ within the baseband $(-\pi/T, \pi/T)$. That is to say, in addition to $\omega = 0$, there will be energy at other $\omega_l = (2\pi/MT)l (l = [-(M/2) : -1, 1 : (M/2)])$, corresponding to the ghosting of the BP image. Then, the expression at $(x_i + q\Delta x_{\text{bp}}, y_i)$ of BP image can be written as follows:

$$\begin{aligned} & i\left(x_i + k_1\Delta x_{\text{bp}}, y_i\right) \\ &= \sum_{t_n} \begin{array}{l} xc_1(t_r, t_n)\cdot\delta\left(t_r - \frac{2}{c}R\left(t_n, x_i + k_1\Delta x_{\text{bp}}, y_i\right)\right) \\ \cdot\exp\left(j\frac{4\pi}{\lambda}R\left(t_n, x_i + k_1\Delta x_{\text{bp}}, y_i\right)\right) \end{array} \\ &= Yc_1\left(j2\pi\frac{2v\cdot q\Delta x_{\text{bp}}}{\lambda R_B\left(q\Delta y_{\text{bp}}, q\Delta x_{\text{bp}}\right)}\right) \end{aligned} \tag{29}$$

where $R_B(q\Delta y_{\text{bp}}, q\Delta x_{\text{bp}}) = ((R_B + q\Delta y_{\text{bp}})^2 + (q\Delta x_{\text{bp}})^2)^{1/2}$, and Δx_{bp} and Δy_{bp} are the range and azimuth image intervals after BP processing. The BP is the ideal state of the PFA, and its essence is to reconstruct the target point spread function (PSF) in the Doppler domain. Through this formula (29), the direct relationship between BP image and Doppler component f_{bp} can be effectively established, which creates conditions for the realization of subsequent BP-domain reconstruction

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TABLE I
RADAR PARAMETERS FOR THE 1-D SIMULATION

Carrier frequency (GHz)		9.6	
Velocity (m/s)	100	PRF (Hz)	2500
M	8	Vertical distance (km)	10

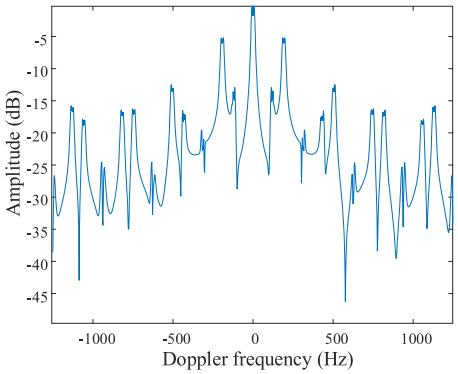


Fig. 9. Doppler spectrum result of the 1-D nonuniform signal.

algorithm, and when $(2v \cdot q\Delta x_{\text{bp}}/\lambda R_B(q\Delta y_{\text{bp}}, q\Delta x_{\text{bp}})) = (l/T_M)$, i.e.,

$$q\Delta x_{\text{bp}} = \frac{l \cdot R_B\left(q\Delta y_{\text{bp}}, q\Delta x_{\text{bp}}\right) \cdot \lambda}{2 \cdot v \cdot T_M} \quad (l \in Z) \tag{30}$$

the ghosting energy will appear near the location $(x_i + q\Delta x_{\text{bp}}, y_i)$.

Equations (29) and (30) show that azimuth grating lobes are inevitable in BP image when SAR signals are periodically nonuniformly sampled in the azimuth direction. The position of the grating lobe can be calculated by (30). To verify this, here is a simple simulation of using BP algorithm to process the 1-D nonuniform sampling signal, and the related parameters are listed in Table I. In this application, the oversampling strategy is employed, and the irregular signal grating lobe energy is dispersed but still exists, as shown in Fig. 9. Besides, the target energy in the center of the scene is stronger, and the ghost target energy generated by it is stronger, as shown in Fig. 10, which is no different from the real target signal. In this case, if the proposed algorithm is not used to suppress the strong ghost target, the overall image quality will also be affected. The correctness of (30) can be verified by calculating the distance between the two ghostings and the ideal point, i.e., $(\pm 1 \cdot R_B \cdot \lambda/2 \cdot v \cdot (M/\text{PRF})) = \pm 488$, as shown in Fig. 10. In Fig. 10, ghostings 1 and 4 are caused by the energy accumulation of the real target at 0 m, and ghostings 2 and 3 are caused by the energy accumulation of the real target at 300 and -300 m, respectively.

Combining (3) and (20), we can further obtain

$$xc_1(t_r, t_n) = \sum_{m=0}^{M-1} \sum_{k=-\infty}^{\infty} xc(t_r, t - (k \cdot M + m)T - r_m T). \tag{31}$$

From (31), it is observed that the data $xc_1(t_r, t_n)$ arrangement is essentially a construction of multichannel model. After the subchannel data division, the expression of the mth

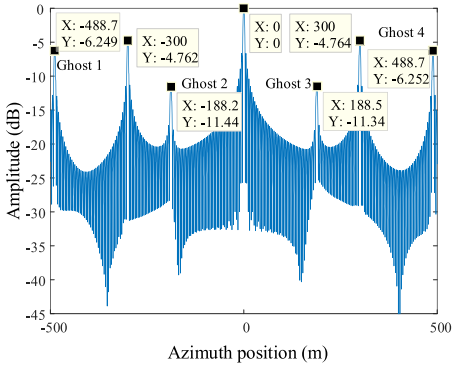


Fig. 10. BP imaging result of the 1-D nonuniform signal.

subchannel signal can be obtained as follows:

$$xc_{1,m}(t_r, t_s) = xc(t_r, t_s - (r_m + m)T) \tag{32}$$

where t_s is the azimuth sampling time obtained after the subchannel data are divided, the sampling period is MT, and the corresponding sampling frequency is $\text{prf}_d = 1/(MT)$.

For each subchannel data, the signal is uniformly sampled in the azimuth direction, but there may be scene Doppler ambiguity, that is, $B_{\text{inst}} > \text{prf}_d$. The specific expression can be expressed as follows:

$$B_{\text{inst}} \approx \frac{2v}{\lambda R_B}(X_{\max} - X_{\min}) > \text{prf}_d \tag{33}$$

where B_{inst} represents instantaneous Doppler bandwidth (IDBW) of the scene. Consider that the azimuth position of the scene center is 0, and $X_{\max} + X_{\min} = 0$.

In order to obtain the image of the whole valid scene, the BP grid layout needs to meet the following condition: $(Q \cdot \Delta x_{\text{bp}}) \geq (2X_{\max})$, where Q is the total number of the BP grids in the azimuth direction. Furthermore, we can have

$$\frac{2v}{\lambda R_B}\left(Q \cdot \Delta x_{\text{bp}}\right) > \text{prf}_d. \tag{34}$$

For uniform sampling, with reference to (30), we can further obtain that the position of the ghosting is

$$q\Delta x_{\text{bp}} = \frac{l \cdot R_B \cdot \lambda}{2 \cdot v}\text{prf}_d. \tag{35}$$

That is to say, the ghostings will appear at $(x_i + q\Delta x_{\text{bp}}, y_i)$. Next, we need to determine whether the ghostings will exist in the valid scene. However, when x_i satisfies the following conditions:

$$\frac{\text{prf}_d}{2} < \frac{2v}{\lambda R_B}|x_i| < \text{prf}_d. \tag{36}$$

Then, the absolute value formula is expanded and discussed in two cases: one case when $x_i < 0$, i.e., $-\text{prf}_d < (2v/\lambda R_B)x_i < -(\text{prf}_d/2)$, we have

$$\begin{aligned} x_i + q\Delta x_{\text{bp}} &= x_i + \frac{(1) \cdot R_B \cdot \lambda}{2 \cdot v}\text{prf}_d \\ &< \frac{\text{prf}_d}{2}\frac{(1) \cdot R_B \cdot \lambda}{2 \cdot v} < X_{\max}. \end{aligned} \tag{37}$$

It can be found that the ghostings will appear in the valid scene, and the position is $x_i + q\Delta x_{\text{bp}} \in (x_i, X_{\max})$. The other

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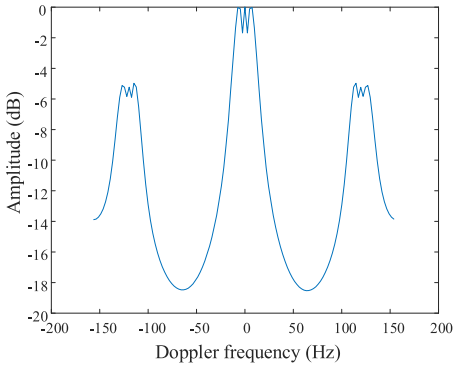


Fig. 11. Doppler spectrum result of the 1-D subchannel signal.

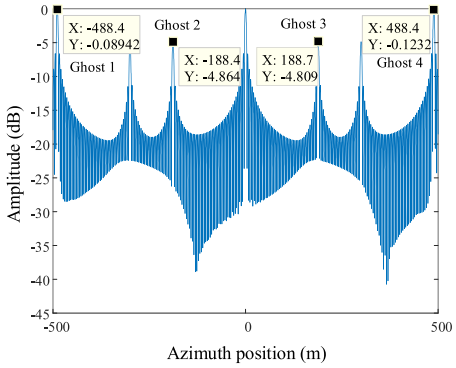


Fig. 12. BP imaging result of the 1-D subchannel signal.

case when $x_i > 0$, i.e., $(\text{prf}_d/2) < (2v/\lambda R_B)x_i < \text{prf}_d$, we have

$$-X_{\max} < \frac{\text{prf}_d}{2}\frac{(-1)\cdot R_B\cdot\lambda}{2\cdot v} < x_i + \frac{(1)\cdot R_B\cdot\lambda}{2\cdot v}\text{prf}_d < x_i + q\Delta x_{\text{bp}}. \quad (38)$$

Note that the ghostings also exist in the SAR scene, and the location is $x_i + q\Delta x_{\text{bp}} \in (-X_{\max}, x_i)$.

In summary, when the subchannel signal has scene Doppler ambiguity, the ghostings will inevitably appear in the subchannel BP image. In order to verify this point, the above 1-D simulation is further performed. Fig. 11 gives the Doppler spectrum results corresponding to one of the subchannel signals after the subchannel division, which is folded in the Doppler domain. Due to the Doppler scene folding of the signal, four ghostings can be clearly observed in the direct result of BP processing, as shown in Fig. 12.

It can be observed that there is a great similarity between the results of Figs. 10 and 12. This is actually due to the structure of $f_{r_m}(\omega)$, which is consistent with the analysis of Section II-A. In other words, Fig. 12 is the BP image corresponding to $Xc_2(t_r, j\omega)$ when $f_{r_m}(\omega)$ does not exist, while Fig. 10 is the BP image result corresponding to $Xc_1(t_r, j\omega)$ when $f_{r_m}(\omega)$ exists. Obviously, the result is the same as that analyzed in the second section. Obviously, it can be seen from the results that the energy of the ghostings in Fig. 10 is weaker than that in Fig. 12, that is, the grating lobe phenomenon in $Xc_1(t_r, j\omega)$ is weaker than that in $Xc_2(t_r, j\omega)$.

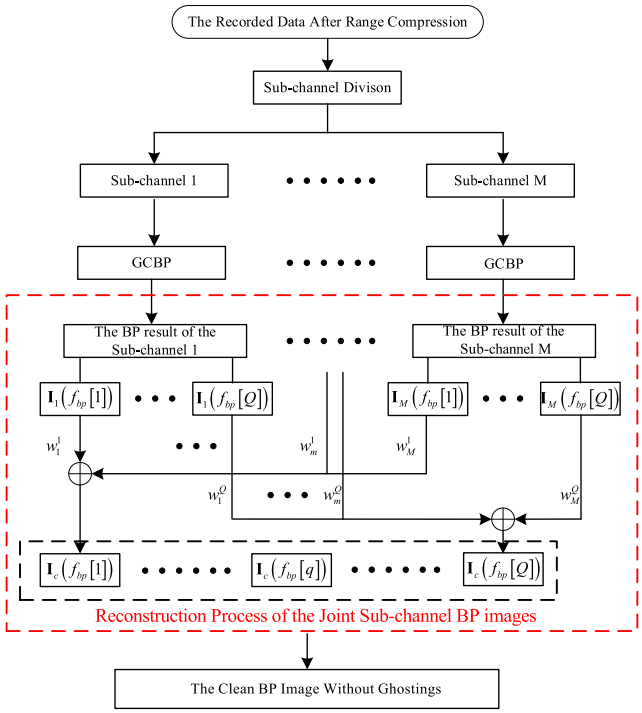


Fig. 13. Flowchart of the proposed reconstruction of the joint subchannel BP image.

III. Proposed Algorithm

A. Flowchart of the Reconstruction of the Joint Subchannel BP Image

The proposed method in this article can be combined with the GCBP algorithm smoothly, so the imaging efficiency can be effectively guaranteed. Besides, the subchannel BP images obtained based on GCBP imaging processing are jointly reconstructed to obtain the imaging results of ghosting-free targets. Fig. 13 shows the whole flowchart of the proposed reconstruction of the joint subchannel BP image. This algorithm mainly consists of three parts: the first step is the division of the nonuniform signal, whose main purpose is to achieve the acquisition of the uniform subchannel signals; the second step is to complete the BP imaging processing of the subchannel signal separately; the third step aims to suppress the ghostings by the reconstruction process of the joint subchannel BP image.

Fig. 14 gives the imaging result after the reconstruction of the joint subchannel BP Image. Compared with the results of Figs. 10 and 12, it can be seen that the ghostings are well suppressed, while the real target is well preserved.

It should be pointed out that when the Doppler bandwidth of nonuniform signal exceeds the $1/(T)$, the traditional reconstruction method [26], [27] performed in the signal domain can only suppress the grating lobes in the baseband $(-1/(2T), 1/(2T))$, and those grating lobes outside the baseband will still be retained. However, the proposed algorithm in this article is based on the joint subchannel GCBP images, the Doppler spectrum of the signal is compressed after the BP processing, and the influence of the signal spectrum aliasing is skillfully avoided. That is to say, both the real target

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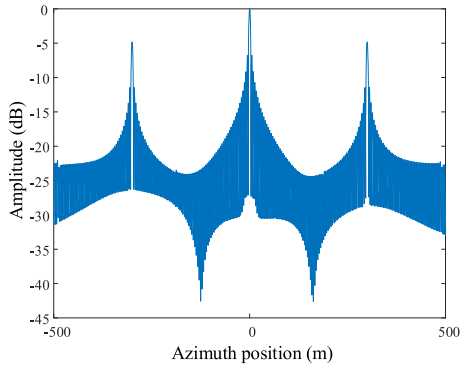


Fig. 14. Imaging result after the reconstruction of the joint 1-D subchannel BP images.

and the ghost are focused on the image domain after BP process, which means that the ghost shadow and the real target are compressed. However, the BP process is a linear process, which will not damage the inherent structure of SAR signal. Therefore, the reconstruction of the joint subchannel BP images can effectively suppress the ghost target while retaining the real target.

B. Reconstruction Model in the Joint Subchannel BP Images

First, the BP process of the subchannel signal is applied, and the expression of the mth subchannel image can be expressed as follows:

$$\begin{aligned}\mathbf{I}_m(\mathbf{x},\mathbf{y}) &= \left[i_m(\mathbf{x},y_1),\ldots,i_m\left(\mathbf{x},y_p\right),\ldots,i_m(\mathbf{x},y_P)\right]^T\\ i_m\left(\mathbf{x},y_p\right) &= \left[i_m(x_1,y_1),\ldots,i_m\left(x_q,y_p\right),\ldots,i_m\left(x_Q,y_P\right)\right]\end{aligned} \tag{39}$$

where Q and P are the number of azimuth and range sampling points in BP image, respectively. Combined with the above analysis and (29), the direct relationship between BP image and Doppler component f_{bp} can be further obtained, i.e., $f_{\text{bp}} = (2v \cdot x_{\text{BP}}/\lambda R_B)$, and $x_{\text{BP}} \in [-(Q/2):(Q/2)] \cdot \Delta x_{\text{bp}}$. Then, we can further give the Doppler component corresponding to the ghosting appearing at $(x + k_1\Delta x_{\text{bp}}, y_i)$ as follows:

$$\frac{x + k_1\Delta x_{\text{bp}}}{2v/(\lambda \cdot R_B)} = \left(f_{\text{bp}} + l \cdot \text{prf}_d\right) \quad (l \in Z). \tag{40}$$

The existence of $l \cdot \text{prf}_d$ means the introduction of the Doppler ambiguity. In other words, in the BP image, the accumulated energy of the target located at $(x + k_1\Delta x_{\text{bp}}, y_i)$ is composed of multiple ambiguous signal components, which can be expressed as follows:

$$\mathbf{I}_m\left(f_{\text{bp}}\right) = \sum_l \begin{array}{l} Yc_1\left(f_{\text{bp}} + l \cdot \text{prf}_d\right)\\ \exp\left(-j2\pi\left(f_{\text{bp}} + l \cdot \text{prf}_d\right)(r_m - r_1)T\right).\end{array} \tag{41}$$

Then, we further have

$$\begin{aligned}\mathbf{\Gamma}\left(f_{\text{bp}}\right) &= \left[\mathbf{I}_1\left(f_{\text{bp}}\right),\ldots,\mathbf{I}_m\left(f_{\text{bp}}\right),\cdots\mathbf{I}_M\left(f_{\text{bp}}\right)\right]^T\\ &= \sum_l Yc_1\left(f_{\text{bp}} + l \cdot \text{prf}_d\right)\mathbf{A}\left(f_{\text{bp}} + l \cdot \text{prf}_d\right).\end{aligned} \tag{42}$$

This formula represents the core multichannel model in the BP image domain in the proposed algorithm, where

TABLE II
RADAR PARAMETERS FOR THE SIMULATION

Carrier frequency (GHz)	11	Squint angle (degree)	0
Velocity (m/s)	180	PRF (Hz)	1200
Bandwidth (MHz)	120	Slant distance (km)	20
Sampling rate (MHz)	144	M	5

$\mathbf{A}(f_{\text{bp}} + l \cdot \text{prf}_d)$ is the BP image-domain steering vector, and the specific expression of $\mathbf{A}(f_{\text{bp}} + l \cdot \text{prf}_d)$ is

$$\mathbf{A}\left(f_{\text{bp}} + l \cdot \text{prf}_d\right) = \left[1,\ldots,e^{-j2\pi\left(f_{\text{bp}} + l \cdot \text{prf}_d\right)(r_m - r_1)T},\cdots\right]. \tag{43}$$

It should be noted that different targets introduce different ghost ambiguity areas in the BP scene. In order to suppress ghostings more effectively, it is necessary to calculate the image-domain steering vectors corresponding to different targets.

C. Joint Reconstruction Processing

By carefully observing (42), it is clear that only by retaining the ambiguity number 0, that is, the scene energy of the corresponding scene area when $l = 0$, the ghosting-free BP result can be obtained, that is, $\mathbf{I}_c(f_{\text{bp}}) = Yc_1(f_{\text{bp}})$. Then, the reconstruction process of subchannel joint BP image can be expressed as follows:

$$\begin{aligned}\mathbf{I}_c\left(f_{\text{bp}}\right) &= \mathbf{W}\left(f_{\text{bp}}\right)\mathbf{\Gamma}\left(f_{\text{bp}}\right)\\ &= \left(\mathbf{A}^+\left(f_{\text{bp}}\right)H_l\right) \cdot \mathbf{\Gamma}\left(f_{\text{bp}}\right)\end{aligned} \tag{44}$$

where "+" is a pseudo-inverse symbol and H_l is the unit matrix.

IV. EXPERIMENTAL RESULTS

In this section, simulation experiments and measured experiments are applied to prove the effectiveness of the proposed algorithm. Furthermore, to prove the good performance of the proposed algorithm, the processing results of the filter group algorithm are added to the following experimental results for comparison.

A. Simulation Data in Nonuniform SAR

In this section, the simulation experiment is carried out, in which the SAR works in the spotlight mode. In the simulation, the data are collected by periodic nonuniform sampling, and the accumulation time is 5 s. Fifteen targets are arranged in the scene, whose size is 400 m in the range direction and 1600 m in the azimuth direction. Table II gives the detailed parameters of the simulations.

The total Doppler bandwidth is 1386 Hz, that is more than a PRF, which is composed of the IDBW of 792 Hz and a signal accumulated Doppler bandwidth of 546 Hz. Fig. 15 gives the Doppler spectrum of the nonuniform signal, where one can see clearly that the spectrum aliasing and grating lobes exist. Then, the BP algorithm is applied for the nonuniform signal, and the

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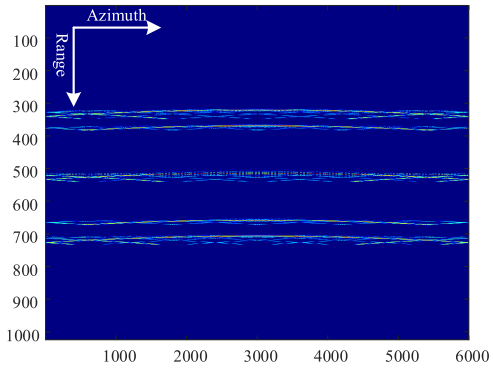


Fig. 15. Doppler spectrum of the nonuniform SAR signal.

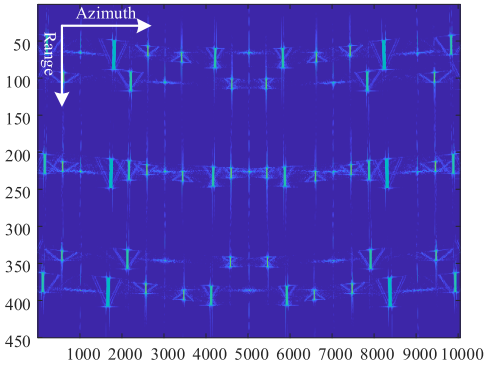


Fig. 16. BP imaging result of the nonuniform SAR signal.

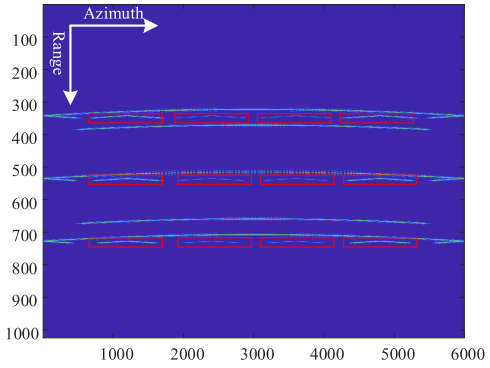


Fig. 17. Doppler spectrum result processed by the filter group algorithm.

imaging result is shown in Fig. 16. However, the existence of ghostings can be clearly seen in Fig. 16, which greatly limits the imaging quality.

Fig. 17 shows the Doppler spectrum result processed by the filter group algorithm, in which the areas marked by the red box are the grating lobes. One can see that the grating lobes in the baseband are well suppressed, while the grating lobes outside the baseband still exist. Furthermore, Fig. 18 shows the BP imaging result processed by the filter group algorithm. It can be seen that most of the ghost targets are well suppressed, but there are still some ghost targets.

Fig. 19 gives the ghosting-free image after the reconstruction of the joint subchannel BP images. At this time, the ghostings have been well suppressed, and the image quality has been greatly improved. By comparing the results of

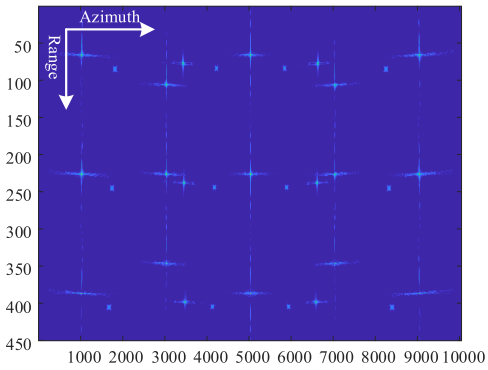


Fig. 18. BP imaging result processed by the filter group algorithm.

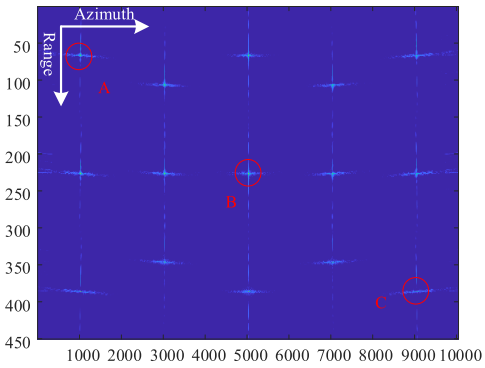


Fig. 19. Ghosting-free image after the reconstruction of the joint subchannel BP images.

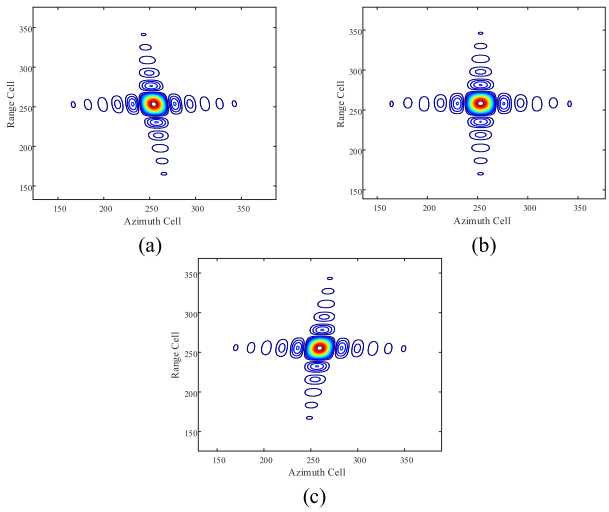


Fig. 20. Two-dimensional contour maps of the three targets (a)–(c).

Figs. 18 and 19, it can be concluded that the performance of the proposed algorithm is better than that of the filter group algorithm when Doppler is aliasing.

Then, the 2-D contour results of the marked target points of Fig. 19 are given in Fig. 20 to further demonstrate the good imaging performance of the proposed algorithm. Therefore, we can conclude that the whole process will not have a bad effect on the focusing state of the real target.

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TABLE III
RADAR PARAMETERS OF THE DATA

Carrier frequency (GHz)	8.9	Squint angle (degree)	0
Velocity (m/s)	100	PRF (Hz)	2000
Bandwidth (MHz)	150	Slant distance (m)	7650
Sampling rate (MHz)	250	M	3

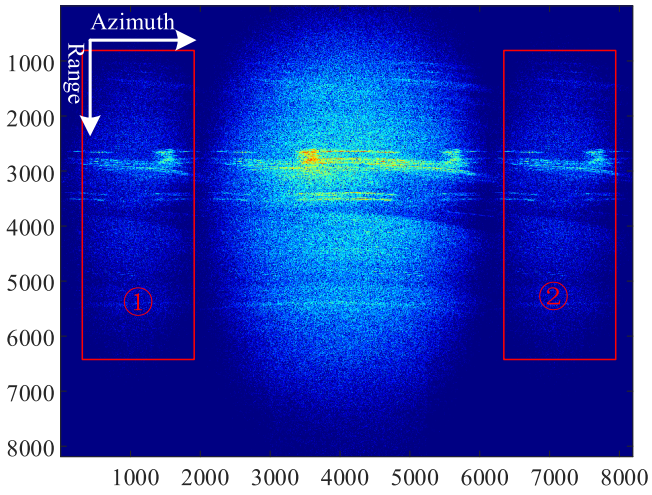


Fig. 21. Doppler spectrum of the nonuniform measured SAR signal.

B. Measured Data in Nonuniform SAR

The experiment based on measured data processing is carried out in this section to prove the effectiveness of the proposed method. The key SAR parameters of this data are listed in Table III. According to the radar parameters, we set the 2-D size of the BP grid to 0.6 and 0.5 m in the range and azimuth, respectively.

Fig. 21 shows the Doppler spectrum of the nonuniform signal, in which the grating lobes can be clearly seen in the marked areas ① and ②. Due to nonuniform sampling, the azimuth grating lobe energy is also relatively strong, which is not much different from the main lobe energy. Then, Fig. 22 shows the BP imaging result, in which the ghostings can be seen in the red frame marker areas. It should be noted that since SAR imaging processing is a strict 2-D gain processing process, the phase history of the signal corresponding to the azimuth grating lobe has a poor match with the matched filter, which leads to defocusing in the ghost area.

Fig. 23 gives the Doppler spectrum of the subchannel signal after subchannel signal division, and the Doppler spectrum of the subchannel signal is aliasing. Fig. 24 shows the BP imaging result of the subchannel signal, in which the existence of ghostings can also be clearly observed. Since the total Doppler bandwidth of the signal corresponding to the measured data is within a PRF, the performance of the filter group algorithm is close to that of the proposed method. Fig. 25 gives the imaging result processed by the filter group algorithm, in which one can see that the ghosting phenomenon is well suppressed. Fig. 26 gives the imaging result after the reconstruction of the joint subchannel BP image, and the ghostings have been well suppressed.

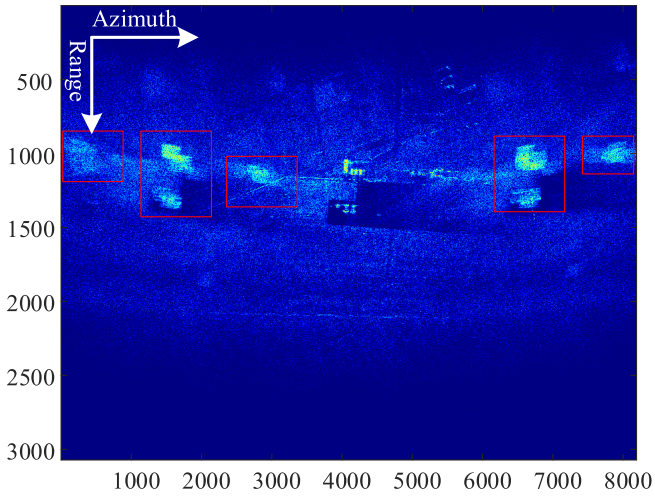


Fig. 22. BP imaging result of the nonuniform measured SAR signal.

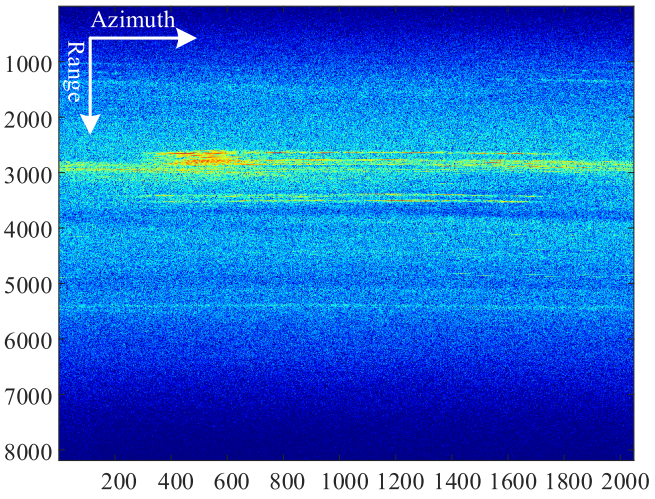


Fig. 23. Doppler spectrum result of the subchannel measured SAR signal.

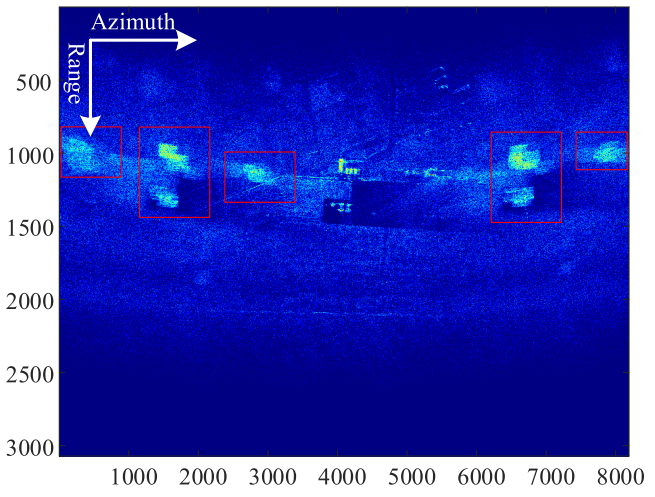


Fig. 24. Subchannel measured SAR imaging result processed by the BP.

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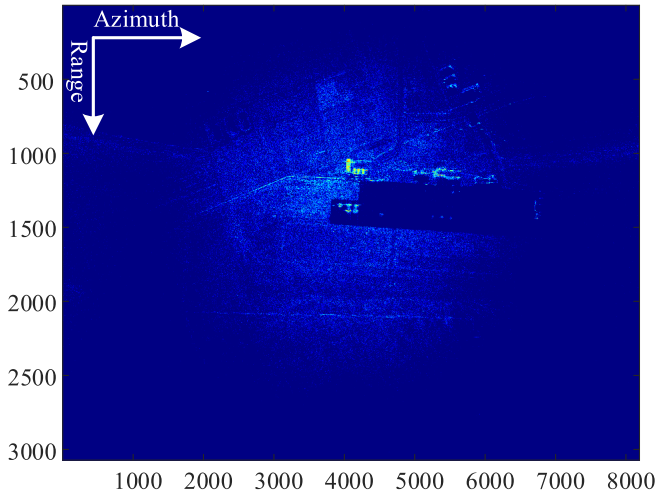


Fig. 25. BP imaging result processed by the filter group algorithm.

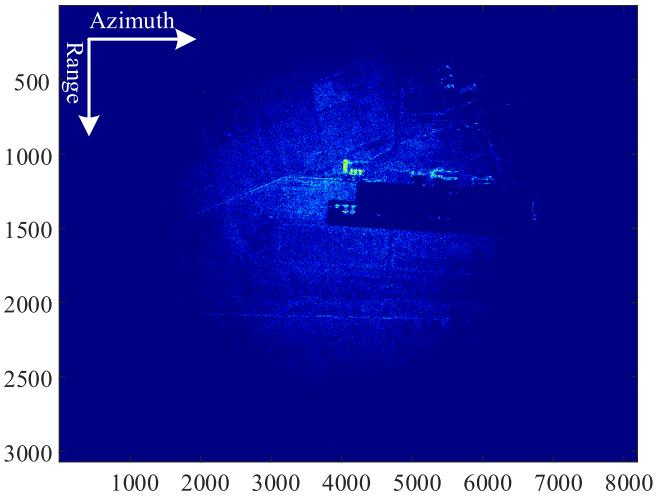


Fig. 26. Ghosting-free image after the reconstruction of the joint subchannel BP images.

V. Conclusion

The proposed algorithm in this article aims to address the issue of ghosting introduced by nonuniform sampling when applying the original BP algorithm. The advantages of the proposed algorithm are as follows.

1) *Wide Applicability:* The algorithm is suitable for arbitrary trajectory imaging processing based on an SAR system.
2) *High Timeliness:* The reconstruction process of the proposed algorithm can be smoothly combined with the GCBP algorithm, ensuring efficient and timely processing.
3) *Better Reconstruction Performance:* The proposed algorithm effectively avoids ghosting introduced by the grating lobes outside the baseband.

To validate the effectiveness of the proposed algorithm, SAR simulation experiments as well as measured experiments are conducted. These experiments provide empirical evidence and demonstrate the capability of the algorithm in suppressing ghosting and improving SAR imaging quality. Overall, the proposed algorithm offers significant advantages in terms of its wide applicability, timeliness, and effectiveness in eliminating ghosting effects caused by nonuniform sampling.

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